

UNIVERSIDADE DO VALE DO ITAJAI – CIENCIAS DA COMPUTAÇÃO

CÁLCULO NUMÉRICO

ALUNO: EDUARDO SLOMP ARÁN

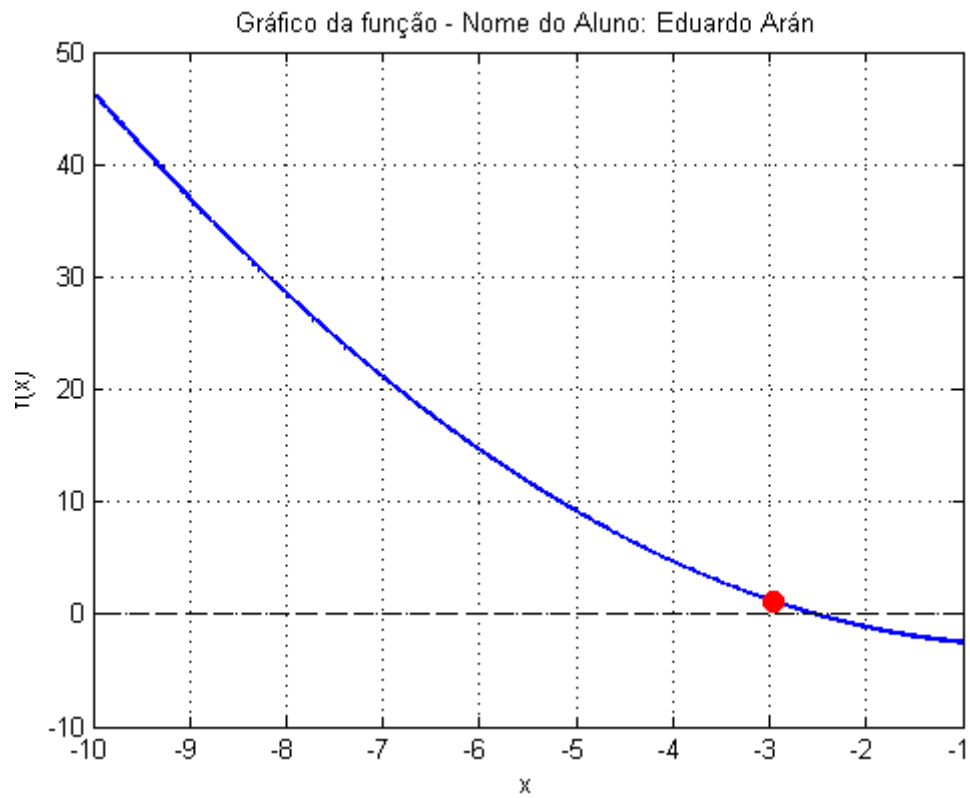
TRABALHO 01

a)

```
function [ fx ] = fbi( x )  
    fx = -x.^(1/3) + 0.5*x.^2 - 2.5;  
end
```

b)

c)



d)

e)

```
>> [raiz, i] = bissecao(a, b, delta)
```

```
raiz =
```

```
-2.9687
```

```
i =
```

```
19
```

DIARY:

clc

a = -10

a =

-10

b = -1

b =

-1

delta = 0.00001

delta =

1.0000e-005

[raiz, i] = bissecao(a, b, delta)

fx =

-6.4402

fx =

-0.1585

o intervalo [a, b] não é correto, mude-o

Error in <a href="matlab:helpUtils.errorDocCallback('bissecacao',
'C:\Users\7975082\Documents\MATLAB\bissecacao.m', 2)" style="font-
weight:bold">bissecacao (<a href="matlab:
opentoline('C:\Users\7975082\Documents\MATLAB\bissecacao.m',2,0)">line 2)

fa = fbi(a); % f(x) avaliado em x=a

{ Output argument "c" (and maybe others) not assigned during call
to "C:\Users\7975082\Documents\MATLAB\bissecacao.m>bissecacao".

}

[raiz, i] = bissecacao(a, b, delta)

raiz =

-2.9687

i =

19

x = linspace(a, b, 1000);

y = fbi(x);

figure;

plot(x, y, 'b-', 'LineWidth', 2)

{ Warning: Imaginary parts of complex X and/or Y arguments
ignored}

plot(x, y, 'b-', 'LineWidth', 2); hold on;

{ Warning: Imaginary parts of complex X and/or Y arguments
ignored}

plot(x, zeros(size(x)), 'k--');

plot(raiz, fbi(raiz), 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');

{ Warning: Imaginary parts of complex X and/or Y arguments
ignored}

```
grid on;
```

```
title('Gráfico da função - Nome do Aluno: Eduardo Arán');
```

```
xlabel('x');
```

```
ylabel('f(x)');
```

```
diary off
```